FPGA Based Computer Vision for Ballistics Localization

Sponsor: N/A

Background: Infantry who come under fire often cannot tell where shooting originates from. The goal of this project is to offer a drone-based resource that can determine the general direction placement where ballistics may originate. Muzzle flashes are bright and easy to spot in an open context but are often obscured in urban settings. This system would watch for muzzle flashes, bullets in travel/impact, and hot gas expelled from barrel using a variety of cameras processed via FPGA fast enough to calculate locations and report to the drone pilot in real time.

Project Needs:

1. Cameras that can be tuned to specific wavelengths with something like a filter. Thermal capabilities for hot gases if a muzzle flash isn't viewable. Standard camera for in-flight controls and bullet trajectory.
2. FPGA processing in real time for each of the cameras and the ability to detect bullet trails. The ability to have at least two sets of camera data to agree before sending information to the pilot.
3. Calculate origin of shot through trajectory, reconstruct current map of area with shot for indoor use, and estimation of accuracy of origin.
4. Video data-based transfer to pilot with overlay of bullet source, direction, friendly/foe, alerts.

Deliverables:

1. A functional static prototype (without drone) using the camera package, FPGA processing, correct bullet tracking/location, frontend video transfer.
2. Documentation of design decisions with rationale of iterations, pivots, and software/hardware choices
3. Record bugs, success/failure of detection rate, metadata such as processing speed, data transfer rate, location accuracy.

Assumptions and Dependencies:

TBD